Ameya S Moghe

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M.tech VLSI and Nanoelectronics
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.tech.	Indian Institute of Technology, Guwahati	8.67 (Current)	2025-Present
B.Tech.	Bhilai Institute of Technology Durg	8.96	2020-2024
Senior Secondary	CBSE Board	90.6%	2020
Secondary	CBSE Board	92.8%	2018

Experience

- **Indian Institute of Technology Guwahati** *Jul 2026 - Nov 2026*
  Teaching Assistant, Digital Circuits Lab *Jan 2026 - May 2026*
  Teaching Assistant, Control and Instrumentation Lab

Projects

- **Software Hardware Codesign For Image Transformer** *Jul 2026 - Present*
  Mtech Project under Prof. Dr. Sheikh Rafi Ahamed,Dept. of EEE IIT Guwahati
  – Dividing the Image Transformer architecture into software tasks and dedicated hardware blocks, and developing RTL code for the hardware blocks with data transmission tested via standard bus protocols.
  – Profiling the Image Transformer's attention and matrix-multiplication layers to identify the highest-impact stages for hardware offload.
- **Design and Layout of Two-Stage Miller-Compensated Op-Amp** *Oct 2025 - Nov 2025*
  Course Project under Prof. Dr. Roy Paylii,Dept. of EEE IIT Guwahati GitHub Repo
  – Designed and simulated a two-stage Miller-compensated CMOS op-amp in **Cadence Virtuoso**, achieving 60.83 dB DC gain and 146.2 MHz unity-gain bandwidth post-layout.
  – Completed full-custom transistor-level layout and verified DRC/LVS-clean; re-simulated with extracted parasitics (PEX) to quantify layout-induced performance loss.
  – Characterized slew rate (87.1 V/µs post-layout), CMRR (≈71 dB), and PSRR (≈56 dB) across pre- and post-layout corners.
- **Design and Analysis of Telescopic Cascode Op-Amp** *Jul 2026 - Aug 2026*
  Self Project GitHub Repo
  – Designed a single-stage telescopic cascode CMOS op-amp in **Cadence Virtuoso**, achieving ≈78 dB DC gain through double cascoding of the input and load devices.
  – Characterized CMRR (≈120 dB) and PSRR (≈40 dB at DC), demonstrating the rejection benefits of the fully cascoded topology.
  – Measured unity-gain bandwidth (≈13 MHz) and slew rate (≈0.99 V/µs) through AC and transient simulation.
- **Design and Analysis of CMOS Bandgap Voltage Reference** *Jul 2026 - Aug 2026*
  Self Project GitHub Repo
  – Designed a PTAT/CTAT-summing CMOS bandgap reference in **Cadence Virtuoso**, using an opamp-forced diode pair (8:1 emitter-area ratio) to generate a temperature-compensated ≈1.2 V V_{REF}.
  – Verified output stability (60–90 mV drift, −40°C to 120°C) and characterized PSRR (≈−21 dB, DC to 1 MHz) through DC and AC simulation.

Technical Skills

- **Programming /HDL Language**: C/C++,Verilog
- **Scripting Language**: Bash*,Tcl* *(* elementary proficiency)*
- **Operating Systems**: Windows,Linux
- **Tools Used**: AMD Vivado,Cadence Genus,Innovus,Tempus,Voltus,Virtuoso(Schematic/Layout/ADE),Spectre Calibre(DRC/LVS),Synopsys VCS,Verdi
- **Language**: Hindi,English,Marathi

Key Courses Taken

- **Core**: Digital IC Design, Analog IC Design, System Hardware Design, Foundation of VLSI Architecture
- **Electives**: Advance Computer Architecture, Foundation of VLSI CAD

Achievements

- **State Level Basketball Player**
- **Visharad in Vocal Music** from Indrakala Sangeet Vishwa Vidyalaya Khairagarh (IKSVV)

Extracurricular and Leadership

- **Hobbies**: Basketball,Chess,Gardening,Cycling,Singing
- **Elected Department Placement Representative**: VLSI,EEE Department IIT Guwahati (May 2026 - May 2027)